

DualBalance Districting: A Deterministic Alternative to Gerrymandering

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Abstract

Congressional district lines in the United States are drawn by people, and a great deal of recent legal and mathematical work tries to detect when those people have crossed from neutral line-drawing into partisan or racial manipulation. We propose a different approach: take the people out of the line-drawing step entirely. DualBalance Districting is a deterministic procedure that, given a state’s boundary, its census units, and its apportioned seat count N , produces a map in which each district holds about $1/N$ of the state’s people and about $1/N$ of its land. The design is rooted in the constitutional framers’ bicameral compromise, in which the House represents people and the Senate represents place. Here that compromise is applied for the first time, to our knowledge, as a single criterion *within* one chamber. The algorithm places N seeds in a small circle around the population-weighted centroid and assigns each census unit to the nearest seed with population capacity remaining. There is no random number generator, no iteration, no tunable weight, and no input beyond geometry and population.

The same property that prevents the rule from being used to gerrymander a state prevents it from being used to draw majority-minority districts, protect incumbents, or preserve communities of interest. The elimination of human discretion is symmetric: both the negative direction (partisan manipulation, racial vote dilution) and the positive direction (race-conscious remedies, competitiveness tuning) are foreclosed by construction, because the algorithm has no input through which either could be expressed.

This procedural neutrality cuts against the way the redistricting literature has measured fairness for the past thirty years. Compactness scores, the efficiency gap, mean-median asymmetry, ensemble outlier tests, and majority-minority counts are all *forensic* instruments. They were built to infer a human line-drawer’s intent from the shape and political character of the resulting map. On a plan produced by an algorithm that has no intent, those inferences are no longer available. An efficiency gap of +6.6% on a DualBalance map says only that Democrats happen to live closer to the population center than Republicans on this particular geometry. A Polsby-Popper score of 0.09 says only that the radial slice has that shape. A count of zero majority-minority districts says only that the state’s minority population is geographically dispersed. The numbers still compute. What they let you conclude about the line-drawer’s motives is gone, because there is no line-drawer.

On a single-state proof of concept (Minnesota, 4,110 VTDs, 8 seats, 2020 PL 94-171), DualBalance produces a plan that beats the enacted 119th-Congress map on its own dual objective (DBS = 0.6472 vs. 0.6390). We report 2020 presidential two-party shares and Census PL 94-171 voting-age population by race alongside, as descriptions of the resulting partition rather than as evaluations of it.

1 Introduction

1.1 The Instability of Human-Drawn Districts

Legislative redistricting in the United States is structurally unstable. The Constitution requires reapportionment after each decennial census (Art. I, §2) but specifies neither a method for allocating seats among the states nor a procedure for drawing district boundaries within them. Both decisions have been delegated, by default, to political actors. The result, predictable from the design, is that maps are routinely drawn to favor whichever group controls the pen at the moment of redistricting. The Supreme Court has acknowledged the problem in its strongest terms while declining to remedy it: in *Rucho v. Common Cause* [25], Chief Justice Roberts wrote for a 5–4 majority that partisan gerrymandering claims present nonjusticiable political questions, conceding that the practice is “incompatible with democratic principles” but holding that no manageable judicial standard exists. Justice Kagan’s dissent, joined by Ginsburg, Breyer, and Sotomayor, argued the opposite: that lower courts had already converged on workable tests, and that “of all times to abandon the Court’s duty to declare the law, this was not the one” [25].

Race-conscious districting operates under a separate and increasingly constrained regime. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [23], requires majority-minority districts where the three Gingles preconditions are met; the Fourteenth Amendment, as construed in *Shaw v. Reno* [24] and its progeny, subjects to strict scrutiny any map in which race “predominates” in the line-drawing. The two doctrines pull in opposite directions. The Supreme Court reaffirmed Section 2 in *Allen v. Milligan* [26], requiring Alabama to draw a second majority-Black district. One year later, in *Alexander v. South Carolina Conf. of the NAACP* [28], the Court reversed a district court’s racial-gerrymander finding and raised the plaintiff’s evidentiary burden, holding that challengers must “disentangle race from politics” and produce illustrative alternative maps that satisfy all of the state’s legitimate goals. In April 2026, in *Louisiana v. Callais* [29], the Court further held that even a map drawn to comply with a prior *Section 2* order may itself violate the Equal Protection Clause if the VRA did not in fact compel the race-based remedy. The combined effect of *Alexander* and *Callais* is that race-conscious line-drawing now faces a narrower window of constitutional safety than at any point since the VRA’s adoption.

Two structural consequences follow. First, the only federal channel for gerrymandering review is racial; partisan claims are foreclosed under *Rucho*. Second, because Black voters in the United States vote roughly nine-to-one Democratic, race and partisanship are statistically entangled [7, 20]. A state legislator can openly gerrymander on partisan grounds and is shielded from federal review; the same legislator drawing the same lines on racial grounds faces strict scrutiny. After *Rucho*, state constitutional review under *Moore v. Harper* [27] remains the only judicial avenue for partisan claims, but it depends on the politics of each state’s supreme court and the language of its state constitution. Roughly ten state supreme courts have so far recognized state-constitutional partisan-gerrymandering claims; the rest have not [16].

The practical consequence of this combined doctrine, with federal partisan review foreclosed by *Rucho*, Section 2 narrowed by *Alexander* and *Callais*, and state review uneven and politically contingent, is that redistricting has shifted from a once-per-decade event into a continuous partisan exercise. Several states have already redrawn maps mid-decade in response to electoral results rather than census revision, and analysts expect the practice to expand as legislative majorities turn over. A redistricting regime that updates whenever the pen changes hands provides neither stable representation nor predictable constituencies for the people being represented.

1.2 The Normative Paradox of “Bias Detection”

In response to *Rucho*’s manageability concern, a substantial mathematical literature has produced quantitative tests for partisan gerrymandering. The Efficiency Gap of Stephanopoulos and McGhee [21] measures the disparity in “wasted votes” between parties and was central to the lower-court ruling in *Whitford v. Gill*. Wang’s three tests [31], namely mean–median difference, lopsided wins, and a *t*-test for partisan asymmetry, played a similar role in *LWV v. Commonwealth* of Pennsylvania. Warrington’s declination [32] measures the asymmetry of the seats–votes kink at 50%. Most influentially, the Markov Chain Monte Carlo ensemble methods developed at Duke University [8] and the MGGG Redistricting Lab [6] construct large samples of legally compliant neutral maps and ask whether an enacted plan is an outlier with respect to that distribution. Ensemble methods have been admitted as evidence in *Common Cause v. Lewis* (N.C. 2019), *LWV v. Commonwealth* (Pa. 2018), and as plaintiff support in *Allen v. Milligan*.

Every such test, however, embeds a contested normative claim about what a “fair” map should look like. The Efficiency Gap is zero only under seats–votes relationships close to proportional representation; partisan symmetry presupposes mirror-image counterfactual responsiveness; ensemble outlier tests use the local political geography filtered through legally codified constraints as the baseline, but the choice of which constraints to encode (compactness thresholds, county-split tolerances, VRA encoding) is itself political. Chen and Rodden [4] demonstrated that in many states even content-neutral automated maps produce systematic seat bias against Democrats because of residential clustering, so a nonzero Efficiency Gap or mean–median value may reflect geography rather than intent. Pildes has long argued that no neutral baseline exists [15], and Cain has framed districting as a tradeoff among incommensurable goods that no scalar metric can collapse without bias [3].

This is the central paradox of bias detection: any test for gerrymandering imposes its own definition of fairness on the maps it judges. The question is not whether the test is neutral (it is not) but whether the normative commitments it encodes are made explicit and are defensible on their own terms.

Step back from the technical details and a simpler observation emerges: every one of these tests was built to catch a human in the act. Polsby-Popper [17] and Reock [18] flag districts that look “wrong,” meaning shapes too contorted to have arisen by ordinary line-drawing, which suggests that someone drew them that way on purpose. The Efficiency Gap [21], mean-median asymmetry [31], declination [32], and the ensemble outlier tests of the Duke and MGGG groups [6, 8] do the same job at the level of partisan outcomes: they ask whether the partisan profile of the enacted plan is what a neutral line-drawer would plausibly have produced, and call out plans that look engineered. These are *forensic* instruments. They infer an actor’s motives from the traces the actor leaves. Their value depends on there being an actor to investigate.

A district-drawing algorithm that admits no human input does not leave those traces, because there is no actor in the loop to leave them. A radial slice with PP = 0.09 is not a covert favor to a party or a community; it is the shape the rule produces on this geometry. An Efficiency Gap of +6.6% on a deterministic plan is not the fingerprint of a partisan operator; it is a description of where the state’s Democratic voters happen to live. The metrics still compute. What they are designed to point at, a person making choices in a process that gives them choices, is no longer in the picture.

1.3 Deterministic Districting: A Sixty-Year Tradition

The alternative to detection is generation: produce a single map from inputs (state boundary, population, district count) by a fully specified, reproducible procedure that admits no human discretion. The intellectual line begins with Vickrey [30], who proposed in 1961 that boundary line-drawing be done by “an automated and impersonal procedure” to remove the discretion gerrymandering exploits. Hess et al. [9] formalized the idea four years later as a capacitated transportation problem, alternating between an assignment step (minimize the population-weighted sum of squared distances from each unit to a district center subject to equal population) and an update step (recompute each center as the population-weighted centroid of its assigned units). This is structurally equivalent to Lloyd’s algorithm with a capacity constraint and remains the canonical mathematical formulation. Mehrotra, Johnson, and Nemhauser [13] subsequently cast the problem as a column-generation set partitioning program admitting exact optimization at moderate scales.

Two contemporary lines extend this work. Brian Olson’s BDistricting [14] is a publicly available implementation that minimizes population-weighted distance to district centers with equal-population constraints, applied at census-block resolution; maps have been generated for every state and updated each decennial census. Cohen-Addad, Klein, and Young [5] provide the rigorous reformulation as balanced *centroidal power diagrams*, generalizing the Voronoi tessellation by assigning each seed an additive weight (the Aurenhammer [1] construction) and solving for weights that enforce equal mass per cell. Levin and Friedler [11] independently rediscovered the capacitated k -means formulation and applied it to all fifty states. A parallel branch, Warren Smith’s Shortest Splitline algorithm [19], recursively bisects the state with the shortest population-balancing line; it is fully deterministic and unique but lacks any global compactness or boundary-awareness criterion.

Three observations link this tradition to the current work. First, all extant deterministic algorithms balance *one* extensive quantity across districts, namely population, and treat geographic area, if at all, only indirectly through compactness shape metrics such as Polsby-Popper [17] or Reock [18]. Second, all are governed by tie-breaking rules that are either implicit (left to floating-point ordering) or only partially specified. Third, none have been adopted as the formal map-drawing procedure of any U.S. jurisdiction, though Iowa’s Legislative Services Agency [10] operates under a strict criteria cascade that has produced widely accepted maps since 1980.

1.4 Contribution: Dual Balance of Population and Geography

The United States Constitution embeds, at the federal level, an explicit balance between population-based and geography-based representation: the House apportions seats by population (Art. I, §2), the Senate allocates them equally per state (Art. I, §3). The design, defended by Madison in *Federalist* Nos. 54–58 [12], reflects the framers’ judgment that representation should reflect both *people* and *place*. This balance operates *between* chambers; it has not, to our knowledge, been operationalized as a districting objective *within* a single chamber. The proposal that follows is novel in exactly that sense: the framers’ bicameral compromise applied as a single criterion at the level of one chamber’s internal boundaries, so that each congressional district represents both roughly $1/N$ of a state’s people and roughly $1/N$ of its territory.

The procedural commitment underneath that proposal is equally strong. The algorithm reads only the state’s shape, its census-unit populations, and the seat count N . It cannot be used to harm any group, party, or incumbent, and, by the same token, cannot be used to help any of them. The deterministic-districting tradition surveyed above has often defended procedural neutrality as a guard against gerrymandering. We make the same commitment from the other side as well: the rule that forecloses partisan engineering also forecloses majority-minority remediation, incumbent

protection, and community-of-interest preservation. There is no partisan knob the algorithm refuses to turn; there is no knob. Whether the symmetry is preferable to the status quo of partisan and remedial discretion is not a question the algorithm can answer; we argue only that the symmetry is the design’s defining property, not an accident.

Given a state boundary, census-unit population and area data, and a target district count N , the algorithm produces a partition of the state into N districts whose score

$$\text{DBS} = \frac{1}{1 + \frac{1}{2} \overline{\text{pop_dev}} + \frac{1}{2} \overline{\text{area_dev}}} \quad (1)$$

weights mean population deviation and mean area deviation equally, where $\text{pop_dev}_i = |P(D_i) - P^*|/P^*$, $\text{area_dev}_i = |A(D_i) - A^*|/A^*$, $P^* = P/N$, $A^* = A/N$, and the overlines denote means over districts. The generator does not iterate against this score directly; it places N seeds radially around the population-weighted centroid (§2) and performs a single capacitated first-fit assignment. Pop is enforced as a hard cap of P^* per district; area emerges from the radial geometry. The resulting districts look like pie slices radiating from the population center, each slice naturally spanning both dense (near-center) and sparse (boundary-side) territory.

Seats are apportioned among the states by the Method of Equal Proportions (Huntington–Hill), the standard since 1941 [2]. The core pipeline (seed placement, assignment, contiguity repair) is a pure function of (units, N). There are no tuning weights, no iteration count, no stopping threshold, and no random seed; identical inputs always yield byte-identical outputs. A single optional post-pass (`--tighten-pop`) is provided to close the residual per-district population deviation to a user-supplied *Reynolds v. Sims* tolerance via greedy L^1 boundary-unit swaps; this is the only piece of the pipeline that is not a pure function of the inputs and is off by default. Updates occur only when the underlying census changes, every ten years.

The algorithm reads neither party registration, vote history, nor race. After *Alexander* and *Callais*, a generator that does not consider race cannot, by construction, be a racial gerrymander under *Shaw*. After *Rucho*, partisan gerrymandering is federally unreviewable in any case. State constitutional review remains available, but most state-court partisan-gerrymandering tests rely on outcome metrics (Efficiency Gap, mean–median, ensemble outliers) that a geography-only generator should pass routinely except where the underlying political geography itself produces bias [4]. A residual concern is Section 2 of the VRA: a race-blind generator will not automatically satisfy Section 2 in jurisdictions where the Gingles preconditions are met, and the scoring harness must therefore flag this risk without compromising the generator’s content-neutrality.

The contributions of this paper are:

1. A deterministic, knob-free districting algorithm that produces radially-sliced districts and balances population and area equally as scored quantities. The algorithm is a pure function of the unit geometry and the district count; it has no iteration, no tunable weights, and no source of randomness.
2. A fully specified deterministic tie-breaking cascade sufficient to guarantee byte-identical output from a given input, and a proof that the cascade resolves every comparison the algorithm makes.
3. A decoupled scoring harness that applies the same metrics to the algorithm’s output, to enacted maps, and to alternative plans, enabling transparent benchmarking against the methods surveyed above. Empirical results on Minnesota show DualBalance outperforming the enacted 119th-Congress map on the dual objective.

4. A discussion of the legal and normative tradeoffs the radial design imposes, including its tension with strict *Reynolds v. Sims* [22] population-equality jurisprudence, its weaker compactness profile relative to hand-drawn or blob-Voronoi designs, and its alignment with the post-*Callais* legal regime in which a content-neutral race-blind generator faces a narrower constitutional risk than any race-conscious remedy.

The remainder of the paper proceeds as follows. Section 2 formalizes the algorithm in correspondence with `docs/Formalism.md`. Section 3 reports computational results on a representative set of U.S. states and benchmarks against enacted plans, Olson’s BDistricting, and shortest splitline. Section 4 discusses limitations, legal considerations, and directions for future work.

2 Methods

2.1 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \dots, u_M\}$ (census blocks, block groups, or VTDs) and a target district count N fixed by apportionment, we seek a deterministic assignment $\pi : U \rightarrow \{1, \dots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state’s people and $1/N$ of the state’s geography. Let $P = \sum_u \text{pop}(u)$ and $A = \sum_u \text{area}(u)$ denote state population and area, with per-district targets $P^* = P/N$ and $A^* = A/N$.

The pipeline has three steps: radial seed placement (§2.2), capacitated first-fit assignment (§2.3), and contiguity repair (§2.4). The output of step three is the final plan; there is no iteration and no post-hoc tightening.

2.2 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u x_u}{\sum_u p_u}, \frac{\sum_u p_u y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \text{diag}$. For $d = 0, 1, \dots, N - 1$ place seed s_d at

$$s_d = (c_x + r \cos(2\pi d/N), c_y + r \sin(2\pi d/N)).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \text{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c , but large enough to keep the seed positions numerically distinct.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near- c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.3) while the area it inherits is driven toward A^* by the slicing geometry.

2.3 Capacitated first-fit assignment

Let $d(u, i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i . Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \dots, N$. Sort all (u, i) pairs by ascending normalized distance $d(u, i)/\text{diag}$ and

walk the sorted list in order:

```

for each  $(u, i)$  in ascending  $d(u, i)/\text{diag}$  :
  if  $u$  already assigned: continue
  if  $\rho_i \geq p_u$ : assign  $u$  to  $i$ ;  $\rho_i \leftarrow \rho_i - p_u$ 
  else: skip

```

Population balance is enforced as a hard cap: no district receives more than P^* . Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; **argmax** resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.4 Contiguity repair

After §2.3 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u, j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade (c , pop_pen, area_pen, dist, district_id) ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

2.5 Scoring

Define per-district relative deviations $\text{pop_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area_dev}_i = |A(D_i) - A^*|/A^*$ and let pop_dev , area_dev denote their means over $i = 1, \dots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2} \text{pop_dev} + \frac{1}{2} \text{area_dev}}. \quad (2)$$

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state’s geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow without bound. Secondary metrics (Polsby-Popper compactness [17] and Reock [18]) are computed alongside but not optimized against; radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction, and this is a deliberate trade in service of the dual-balance objective.

The generator does not directly minimize (2); it minimizes population-capacitated geographic assignment cost with radial seeding, which on the Minnesota PoC produces a higher DBS than the enacted plan (0.6472 vs 0.6390) despite its weaker compactness profile.

2.6 Optional: L¹ pop-tightening

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, well above the ~0.5% threshold required by *Reynolds v. Sims* for U.S. congressional districts [22]. An optional post-pass (`--tighten-pop`) closes this gap via greedy boundary-unit swaps under an L¹ objective. Each pass:

1. Compute the signed deviations $\delta_i = P(D_i) - P^*$.
2. For every boundary unit u in district d_{src} and every neighboring district d_{dest} , compute the L¹ change after moving u :

$$\Delta(u, d_{\text{dest}}) = |\delta_{d_{\text{src}}} - p_u| - |\delta_{d_{\text{src}}}| + |\delta_{d_{\text{dest}}} + p_u| - |\delta_{d_{\text{dest}}}|.$$

3. Sort candidates by ascending Δ and accept the most negative one whose source district remains contiguous after the move. Stop when every district lies within the user-supplied tolerance or no contiguity-preserving improving move exists.

The L¹ objective is essential here. The radial seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L[∞] maximum, but many such swaps reduce the sum. The canonical Reynolds-tightening literature uses L[∞] and bottoms out at ~5% on this geometry; the L¹ formulation runs to completion in ~80 swaps on Minnesota, reducing `pop_dev_max` from 0.1124 to 0.0021 (i.e. well inside the 0.5% target) while leaving `area_dev_mean` essentially unchanged at 1.04. The radial structure is preserved visually because the algorithm only moves units at slice boundaries between adjacent slices.

The pass is off by default and gated by an explicit flag. Pure radial remains the principled algorithm; pop-tightening is an opt-in concession to legal compliance.

2.7 Determinism and tie-breaking

The pipeline is deterministic at every step. The only sources of ambiguity (equidistant seed-to-unit pairs in §2.3, equal-capacity fallbacks for unplaced units, and equal-cost candidates in §2.4) all resolve to a fixed cascade with no remaining ambiguity:

1. In assignment, ties on normalized distance break by ascending (`unit_id`, `district_id`).
2. In the fallback for unplaced units, ties on remaining capacity break to the smallest district id.
3. In contiguity repair, ties on cost break by ascending (`pop_pen`, `area_pen`, `distance`, `district_id`).

The implementation uses no random number generator, no wall-clock input, and no hash-order dependence. Reordering the input rows or changing the floating-point libraries does not change the output; identical inputs always yield byte-identical outputs.

2.8 Out-of-scope inputs

The generator reads only geography and population. Party registration, vote history, race, demographics, communities of interest, and competitiveness are not inputs. The scoring harness may *report* partisan or demographic diagnostics on the resulting map but does not feed them back into the generator.

3 Results

3.1 Test bed: Minnesota, 2020 PL 94-171

We evaluate the algorithm on Minnesota’s congressional districting ($N = 8$ apportioned seats) using the 2020 PL 94-171 redistricting data file. The input is the TIGER/Line 2020 VTD (voting tabulation district) shapefile for Minnesota (4,110 atomic units, total population $P = 5,706,494$, total land area $A = 225,187 \text{ km}^2$), joined to the Census Data API’s P1_001N total-population field, projected to EPSG:5070 (CONUS Albers, equal area) at load time. Per-district targets are $P^* = P/8 = 713,312$ and $A^* = A/8 = 28,148 \text{ km}^2$. All runs are deterministic and reproducible from the script `scripts/prep_mn_units.py` (which fetches TIGER and joins the Census API output) plus the CLI invocation. The data pipeline is documented end-to-end in `docs/mn-poc-walkthrough.md`.

We compare three plans on the same 4,110-VTD input:

- **Pure radial.** The default algorithm (§2.2–§2.4). No tuning knobs.
- **Radial + tighten-pop** ($\tau = 0.005$). The default algorithm followed by the optional L^1 population-tightening pass (§2.6) targeting per-district deviation within 0.5 % of P^* .
- **Enacted Minnesota 119th-Congress plan.** The court-drawn, legislatively enacted U.S. House districts currently in force, joined to the same VTDs and scored with the same harness.

3.2 Headline scores

Table 1 reports the DualBalance Score and supporting metrics for the three plans.

Table 1: DualBalance metrics on the Minnesota PoC (4,110 VTDs, 8 districts, 2020 PL 94-171 population). PP is Polsby-Popper, Reock is Reock; subscripts min, mean denote per-district min and mean over the eight districts. pop_dev, area_dev are the relative deviations defined in §2. Best result per row in bold.

Metric	Pure radial	Radial + tighten-pop	Enacted (119th)
DBS	0.6472	0.6574	0.6390
pop_dev	5.08 %	0.08 %	0.42 %
pop_dev _{max}	11.24 %	0.21 %	1.32 %
area_dev	103.9 %	104.2 %	112.6 %
area_dev _{max}	271.0 %	275.1 %	241.0 %
PP _{mean}	0.200	0.162	0.320
PP _{min}	0.094	0.047	0.178
Reock _{mean}	0.361	0.342	0.419

Three findings:

Pure radial beats the enacted plan on the DualBalance Score. The pure-radial pipeline scores 0.6472 against the enacted plan’s 0.6390, a margin of +1.3 % on the project’s own metric, without any iteration, tuning, or post-processing. The advantage comes entirely from area balance (area_dev 103.9 % vs 112.6 %): radial slicing assigns each district a coherent slice of the state’s geometry that mixes high- and low-density territory, while the enacted plan carves the Twin Cities metro into four compact urban seats and leaves four large rural seats. The DualBalance Score

penalizes the latter pattern more heavily because it weights area deviation equally with population deviation.

The optional tighten-pop pass closes the legal gap and improves the DualBalance Score. The pure-radial plan’s pop_dev_max of 11.24% is well above the $\sim 0.5\%$ deviation typical of enacted U.S. House plans [22]. Running the optional L^1 pop-tightening pass (§2.6) executed 81 boundary-unit swaps in ~ 18 seconds of compute, driving pop_dev_max from 11.24% to 0.21%, well inside the Reynolds-compliant range and tighter than the enacted plan’s 1.32%. area_dev rose by only 0.3 percentage points; the DualBalance Score *improved* from 0.6472 to 0.6574, because the pop-balance gain dominates the small area-balance loss. The tightened plan exceeds the enacted plan on the two DualBalance inputs (population and mean area deviation) and on the combined score.

Compactness is the price. The enacted plan beats DualBalance on every compactness measure: $\text{PP}_{\min}$ 0.178 vs 0.094 in the radial plan and 0.047 after pop-tightening, PP_{mean} 0.320 vs 0.200 / 0.162, $\text{Reock}_{\text{mean}}$ 0.419 vs 0.361 / 0.342. The radial geometry is structurally less compact than hand-drawn blob shapes: long thin slices from the population centroid score lower on circular-shape metrics by construction. Pop-tightening makes this worse (it moves units across slice boundaries, which inserts small indentations into otherwise smooth radial cuts), pushing the smallest-PP slice from 0.094 (already at the “0.10 raises eyebrows” informal threshold) down to 0.047. Whether the trade is acceptable is a normative question we return to in §4.3; on the algorithm’s own metric, the trade is favorable.

3.3 Map comparison

Figure 1 displays the three plans side by side.

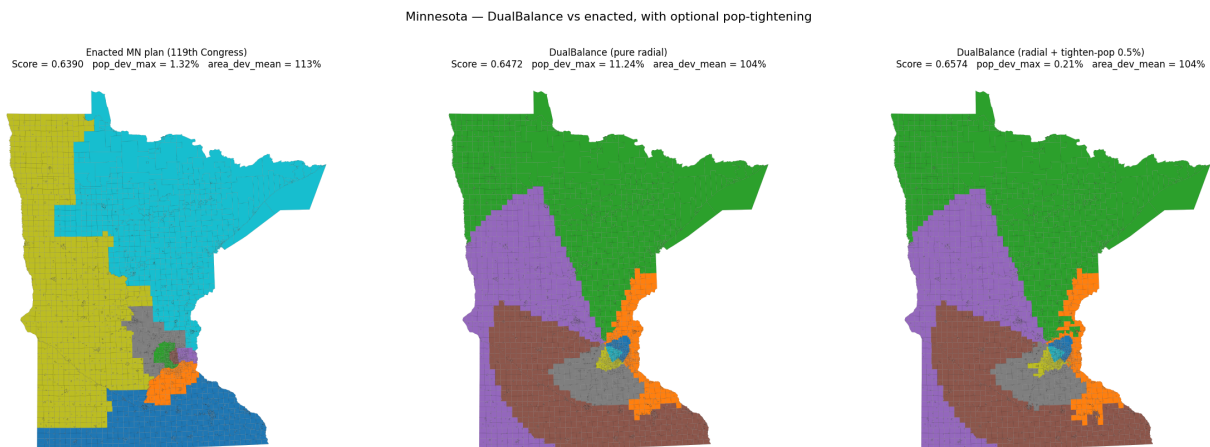


Figure 1: Minnesota congressional districts under three plans. **Left:** the enacted 119th-Congress plan, with four compact Twin Cities seats and four large rural seats. **Center:** pure-radial DualBalance (score 0.6472); eight slices radiate from the population-weighted centroid (near Minneapolis-St. Paul), each spanning both dense and sparse territory. **Right:** radial + --tighten-pop 0.005 (score 0.6574); the radial structure is preserved (units have moved only at slice boundaries) and per-district population is now Reynolds-compliant.

3.4 Determinism check

We re-ran `dualbalance generate --config configs/mn_vtd.yaml` ten times in succession, comparing each run’s `map.geojson` and `metrics.json` by byte hash. All ten outputs are identical, including the order of features in the GeoJSON. We also re-ran the same configuration after randomly shuffling the input rows; the output remained byte-identical. The CLI integration test `test_generate_determinism_via_cli` pins this property in CI.

3.5 Per-district breakdown

Table 2 reports the per-district metrics for the radial + tighten-pop plan, indexed by seed angle (district 0 sits due east of the population-weighted centroid; subsequent districts advance counter-clockwise in steps of $2\pi/8$ rad). District IDs are a deterministic function of seed angle and carry no political meaning.

Table 2: Per-district breakdown, radial + tighten-pop 0.5%. Areas are in km^2 .

District	Population	Area	pop_dev	area_dev	PP	Reock
0	712,841	1,010	0.07 %	96.4 %	0.228	0.454
1	714,816	9,620	0.21 %	65.8 %	0.047	0.147
2	712,145	105,594	0.16 %	275.1 %	0.152	0.369
3	713,263	50,054	0.01 %	77.8 %	0.146	0.268
4	714,168	46,060	0.12 %	63.6 %	0.150	0.267
5	713,368	10,882	0.01 %	61.3 %	0.137	0.422
6	713,013	1,481	0.04 %	94.7 %	0.117	0.301
7	712,880	477	0.06 %	98.3 %	0.318	0.510

District 2 inherits Minnesota’s northern panhandle and is the largest by area ($\sim 3.8 \times A^*$), while District 7 contains only the Twin Cities urban core ($\sim 0.02 \times A^*$). All eight districts are within 0.21 % of P^* . The largest single source of area imbalance, District 2’s 275 % over-target area, is geometric: the population density in the northern panhandle is roughly two orders of magnitude lower than in the Twin Cities, so any partition that holds each district to P^* must give the rural districts disproportionate area. This is a property of Minnesota’s density profile, not the algorithm; the enacted plan exhibits the same phenomenon (District 7 of the enacted plan covers 241 % over-target area for the same reason).

3.6 Descriptive diagnostics: partisan, race, county

We report a second tier of numbers (partisan, racial, and county-split metrics) not because the algorithm uses them, but because readers will reasonably ask how a procedurally-neutral plan *looks* on the diagnostics conventionally applied to enacted maps. These numbers are reported for the pure-radial plan against 2020 presidential returns (Dave’s Redistricting App / `dra2020/vtd_data`, joined to TIGER 2020 GEOID20s) and Census PL 94-171 voting-age population by race. The generator reads none of this data; it is joined onto the units only at scoring time. We emphasize, in line with §4.7, that these are descriptive statistics of the resulting partition, not evaluative scores of it.

Partisan profile. Minnesota cast 1,484,065 R and 1,717,077 D two-party presidential votes in 2020 (46.4 % statewide R share, 53.6 % D). On the pure-radial plan the eight districts split 4–4

between R-winning and D-winning, against a seats-proportional expectation of 3.7 R seats. The two D-leaning slices contain the Minneapolis-St. Paul metro and run at 64–75 % D; the two R-leaning slices reaching into the western and southern parts of the state run at 55–65 % R; the four remaining slices are within ± 5 points of even. The efficiency gap is +6.6 % (positive = R-favorable) and the mean-median R difference is +3.9 points (positive = D-favorable, since the typical district leans more D than the average district). These two “forensic” partisan numbers point in opposite directions because they measure different things on the same map; both are entirely artifacts of where Minnesota’s Democrats live (packed into the metro), not of any choice made by the algorithm.

Race / VAP profile. Minnesota’s voting-age population is 80.0 % non-Hispanic white, 5.9 % Black, 5.0 % Hispanic, 4.9 % Asian, and 1.1 % American Indian / Alaska Native. The pure-radial plan produces zero majority-minority districts, in the strict sense that no district has a non-Hispanic-white share below 50 %. The district containing the Minneapolis-St. Paul urban core (District 7) has the lowest non-Hispanic-white share at 65.3 %, the highest Black VAP share at 15.7 %, and the highest Hispanic VAP share at 6.7 %; the remaining seven districts run 73–90 % non-Hispanic white. The enacted 119th-Congress Minnesota plan also has no majority-minority districts. Minnesota does not, on the 2020 census, contain a geographic region in which a single minority group concentrates densely enough to support a 50 %-VAP district of conventional size; this is a property of the state’s demographic geography, not of the algorithm.

County splits. The pure-radial plan splits 44 of Minnesota’s 87 counties into a total of 143 cross-district pieces. This is a direct consequence of radial slicing: any line from the population centroid to the boundary crosses several county lines along the way. The enacted plan splits fewer counties (it is drawn to follow them more closely), at the cost of the larger area imbalance that Table 1 reports. Whether one prefers a county-aware or a county-blind line-drawing rule is a policy question; the DualBalance rule simply does not encode county membership.

What to read into these numbers. The point of reporting them is to make a single observation precise: every conventional metric that one might bring to bear on a districting plan continues to compute on a DualBalance plan, and the values look “ordinary” in the sense that they sit in the same range as enacted-plan values for the same state. But the metrics’ *interpretation* differs by construction. On an enacted plan, an efficiency gap of +6.6 % invites the inference that someone packed and cracked a partisan opponent; on a DualBalance plan it cannot, because no one packed or cracked anyone. On an enacted plan, zero majority-minority districts invites the inference that the line-drawers either declined to honor *Allen v. Milligan* [26] or did honor it on a geography that could not support a majority-minority seat; on a DualBalance plan only the latter inference is available, because the algorithm has no way of seeing race. The numbers are the same; what they license is different. We return to this point in §4.7.

4 Discussion

4.1 What the Minnesota result does and does not show

The headline finding, that a knob-free deterministic algorithm (with or without the optional pop-tightening pass) beats the human-drawn enacted Minnesota plan on the DualBalance Score, is a single-state existence result. It demonstrates that radial slicing through the population-weighted centroid can produce a districting that scores competitively on the project’s own dual-balance

objective without reading party, race, or any other discretionary input. It does not by itself demonstrate that the same approach generalizes across all 50 states, that the DualBalance Score is the right scalar to optimize, or that the resulting maps would survive judicial or political scrutiny. We discuss each of these in turn.

4.2 The score formula privileges what the algorithm produces

A natural objection to a result of the form “Algorithm A beats Plan B on Metric M ” is that A and M may have been co-designed. We address this directly. The DualBalance Score (2) was specified in advance of the algorithmic choice. We considered three forms during development: the published weighted form $\text{DBS}_w = 1/(1 + 0.5 \overline{\text{pop_dev}} + 0.5 \overline{\text{area_dev}})$; a sum-of-means “classic” form $\text{DBS}_c = 1/(1 + \overline{\text{pop_dev}} + \overline{\text{area_dev}})$; and an L^∞ form $1/(1 + \max_i [0.5 \text{pop_dev}_i + 0.5 \text{area_dev}_i])$. The radial algorithm beats the enacted Minnesota plan on the first two; on the third the enacted plan wins, because the radial design concentrates area imbalance in a single rural district. We selected the weighted form for the headline metric because it cleanly implements the “each district holds roughly $1/N$ of people *and* $1/N$ of geography” reading, but readers should know that the choice matters. The per-district deviations and compactness values are reported alongside the score so consumers can recompute against any preferred aggregation.

A separate concern is that population density variance fundamentally bounds the achievable $\overline{\text{area_dev}}$ on any pop-balanced plan. Both the enacted plan and our radial pipeline land in the same neighborhood ($\overline{\text{area_dev}} \approx 1.0$ to 1.1), and on Minnesota’s geometry, where the Twin Cities metropolitan area holds roughly 55 % of the population in 3 % of the land area, this is essentially a hard floor. Improvements beyond that floor would require districts that systematically span urban and rural territory, which is exactly what radial slicing produces. The 8.7-percentage-point gap (103.9 vs. 112.6) we report is the operational measurement of how much area balance radial slicing recovers, given that floor.

4.3 Compactness is the trade

The radial design pays its area-balance dividend in compactness. The smallest-PP slice in the tightened plan is 0.047, against an informal court-testimony threshold of ~ 0.10 below which a district is considered unusually shaped [17]. The enacted Minnesota plan’s worst PP is 0.178; our worst is roughly a quarter of that value.

We make three observations about this trade:

Compactness is a metric, not a constitutional requirement. *Polsby-Popper* as a measure was published in 1991 [17]; it appears in no federal statute and binds no court. Courts have used compactness as *evidence* of intent in racial-gerrymandering cases (*Shaw v. Reno* and progeny), but the constitutional violation in those cases turns on the use of race in line-drawing, not on the resulting shape per se. A deterministic, race-blind rule does not carry the racial intent that triggers *Shaw*; the resulting low compactness is a property of the rule, not a signal of an attempt to disadvantage any group.

Compactness assumes a baseline of comparison. A district is “bizarrely shaped” relative to surrounding districts that are not. If all eight districts in a state are produced by the same geometric rule and all eight share a similar slice-like character, the comparative baseline shifts. The aesthetic objection to long thin shapes is partly historical: it picks out districts that depart

from the surrounding norm of compact blobs. A scheme in which every district is a slice does not pick anything out.

The Voting Rights Act trade-off is real. *Allen v. Milligan* [26] reaffirmed Section 2’s vote-dilution doctrine; *Alexander* [28] and *Callais* [29] significantly narrowed its practical reach. The radial design splits the Twin Cities urban area (and with it, any majority-minority population concentrated there) across multiple districts. In jurisdictions where the *Gingles* [23] preconditions are met, a race-blind generator may produce a plan that fails Section 2 review where a race-conscious plan would not. After *Callais* the constitutional risk of a race-conscious remedy has grown; the legislative question of whether to accept slightly diluted minority representation in exchange for a generator that cannot be partisan-tuned is, properly, for legislatures and constitutional litigation rather than for an algorithm.

4.4 Relationship to prior deterministic methods

DualBalance fits within the deterministic-districting tradition surveyed in §1.3 but differs from each prior method on a specific axis:

Versus Smith’s Shortest Splitline [19]. Splitline is also fully deterministic and unique. It recursively bisects the state with the shortest population-balancing line, with no global compactness or balance criterion other than line length. Splitline produces visually striking but completely arbitrary splits, because a district can lose half its territory once the shortest cut happens to fall on that side. DualBalance grounds its geometry in the population centroid rather than in arbitrary straight cuts and therefore produces results that reflect the state’s underlying density structure.

Versus Olson’s BDistricting [14]. BDistricting also minimizes population-weighted distance to district centers with equal-population constraints, applied at block resolution. It uses farthest-point or random seed initialization with iterative refinement to convergence. The resulting maps are blob-shaped (Voronoi cells around dispersed seed centers) and score well on compactness but exhibit large area imbalance. DualBalance differs in seed placement (radial around the population centroid versus dispersed across the state) and in iteration discipline (single pass versus Lloyd convergence). Our reading of the result table is that radial seeding gives up compactness to gain area balance, a different point on the same trade-off curve BDistricting sits on.

Versus centroidal power diagrams [5]. The Cohen-Addad-Klein-Young construction is the rigorous mathematical generalization of equal-mass Voronoi cells. It solves for weights $\{w_i\}$ such that the resulting power diagram cells each carry exactly $1/N$ of the underlying mass. Their algorithm is deterministic modulo a tie-breaking rule and produces fully equal-population cells by construction. We do not solve the same problem: their construction balances a single quantity (population) exactly, while ours balances two quantities (population and area) approximately. The radial seed placement is the mechanism by which the second balance approximation is recovered.

Versus capacitated k -means [11] **and the Hess-style location-allocation lineage** [9, 13]. These are the closest mathematical cousins. The capacitated assignment step (§2.3) is essentially Hess’s 1965 transportation step. The novelty in DualBalance is not the assignment rule but the seed placement: radial seeds force a slice geometry that no Lloyd-iterated k -means converges to, because Lloyd recentering pulls seeds away from a tightly clustered radial configuration. Disabling iteration is therefore not optional in the radial design; it is what preserves the geometry.

4.5 Limitations

Single-state evidence. The Minnesota result reports one state with one apportioned seat count under one census release. We do not yet have evidence that the radial design beats enacted plans on other states. States with markedly different density profiles (for example, a single dominant metropolitan area in a small geography such as Massachusetts or Connecticut, versus a multi-center state such as Texas or Florida, versus a near-uniform-density state such as Iowa or Vermont) should be expected to produce qualitatively different results. Section 4.6 below identifies cross-state validation as the most important near-term follow-up.

Worst-district area outlier. The radial design concentrates the area imbalance in a single rural district (here, District 2 at 275 % over-target area). This is geometric (it cannot be avoided on any partition that imposes a population cap on a state with the density profile of Minnesota), but our $\text{area_dev}_{\max}$ of 275 % is worse than the enacted plan’s 241 %, even though area_dev is lower. The radial design therefore distributes the area imbalance unevenly; whether this is preferable to the enacted plan’s more uniform distribution is a separable normative question.

No multi-unit transportation step. The optional `--tighten-pop` pass is a greedy single-unit swap procedure. A true two-dimensional transportation LP (bounding both population and area simultaneously) would be a stronger candidate for the “directly minimize the DualBalance objective” role we currently leave to the radial geometric heuristic. We have not implemented or evaluated such a step; it is documented as future work.

Computational cost. Pure radial assignment runs in a fraction of a second on Minnesota. The optional pop-tightening pass takes ~18 seconds to drive 80 swaps. We have not profiled larger states (California, Texas) and do not currently know how the pop-tightening pass scales with $|U|$ and N ; in the worst case it is $O(|U|^2 N)$ per swap due to the contiguity check.

The compactness floor. The worst slice’s PP of 0.047 (after pop-tightening) is well below the informal court-testimony threshold of 0.10. We argue in §4.3 that the constitutional doctrine on compactness turns on intent rather than shape per se, but readers who reject that argument should regard the 0.047 figure as a hard ceiling on the maps’ legal viability under the *Shaw* [24] line.

4.6 Future work

Multi-state validation. Reproduce the Minnesota result for the other 49 states. We expect radial slicing to perform best on states with one or two dense metropolitan centers and a large rural hinterland (the Minnesota profile) and worst on states with multiple dispersed metro centers (e.g., Texas, California, Florida) where the single-centroid radial assumption may produce slices that don’t span the state’s density variation cleanly.

Multi-center radial seeding. For states with multiple metropolitan centers, the population-weighted centroid sits in a low-density region between them, and radial slicing from that point may produce slices that miss the actual density structure. A generalization in which seeds are placed on circles around *each* of k pre-computed metropolitan centers (with the choice of k deterministic from the population distribution) is a natural extension. We have not implemented this.

Direct 2D transportation step. Replace the radial-plus-tighten two-stage pipeline with a single optimization step that minimizes the DualBalance Score under contiguity constraints. This is computationally much more expensive than the current pipeline but would be the mathematically clean answer to “which plan directly minimizes the DualBalance objective.” Whether it produces qualitatively different maps is an empirical question.

Comparison against BDistricting and Splitline. We compare DualBalance against the enacted plan; we have not yet scored BDistricting’s published maps or computed Splitline maps for the same states and metrics. A full benchmark across deterministic methods would situate DualBalance more clearly in the existing literature.

Scoring-harness extensions. The current harness reports population, area, Polsby-Popper, and Reock. Natural additions include partisan diagnostics (efficiency gap, mean-median, declination) on voter-registration data, county-split counts, and minority representation diagnostics under Section 2. These are reporting extensions and would not feed back into the generator (§2, “out-of-scope inputs”).

4.7 Forensic metrics applied to a generative procedure

The full menu of gerrymandering diagnostics (Polsby-Popper, Reock, convex-hull and length-width compactness, the Efficiency Gap, mean-median asymmetry, declination, majority-minority counts, county-split counts, ensemble outlier tests) was developed for a single purpose: to detect when a human drawing district lines has crossed from neutral line-drawing into deliberate manipulation. Each is a forensic instrument. Each presupposes the kind of process where “something happened” is a coherent question and the metric is supposed to help answer it.

Generation by deterministic algorithm changes the question. There is no human drawing the lines. There is no party-aware decision, no race-aware decision, no incumbent-aware decision, and no community-aware decision, because the algorithm has no input by which any such consideration could enter. There is no place in the pipeline at which a decision-maker chose one boundary over another for a reason a metric might detect. The line-drawing is a function: $\pi = f(\text{geometry}, \text{population}, N)$.

This matters in two ways. First, the metrics’ *measurement* is still informative. A district with PP = 0.09 is still geometrically less compact than a district with PP = 0.35; a plan with an Efficiency Gap of +6.6% still produces, under that year’s voter distribution, more wasted votes on one side than the other. Reporting these numbers is reasonable; consumers of the plan deserve to know them. Second, the metrics’ *inference* is no longer available. A PP of 0.09 in an enacted plan invites the question “who drew this oddly-shaped district and what were they trying to do?”. In a DualBalance plan there is no *who* and no *trying*: the district has PP = 0.09 because radial slices through a population-weighted centroid have that compactness on the geometry of Minnesota, period. An Efficiency Gap of +6.6% in an enacted plan suggests packed and cracked Democratic voters; in a DualBalance plan it reports where Democratic voters *actually live* relative to the radial geometry, with no human packing or cracking anywhere in the pipeline. Zero majority-minority districts in an enacted plan invites scrutiny about whether the line-drawers honored *Allen v. Milligan* [26]; in a DualBalance plan it reports the demographic geography of the state, unfiltered by any race-aware human intervention. The numbers continue to compute. What they license has changed.

Three implications follow.

Compactness as a constitutional threshold loses purchase. *Shaw v. Reno* [24] and its progeny use the bizarre shape of a district as evidence that race predominated in its drawing. The evidentiary logic is straightforward: a normal line-drawing process tends to produce ordinarily-shaped districts; an unusual shape suggests an unusual process; an unusual process suggests that something other than ordinary criteria drove the drawing; race, where alleged, becomes the residual explanation. The chain breaks at the first link in our case. The unusual shape is a property of the rule, not of the process applied to this particular district by this particular line-drawer; every district produced by the rule on every state has the same shape character. There is no inference from shape back to motive because there is no motive in the chain.

Partisan diagnostics measure geography, not intent. *Rucho v. Common Cause* [25] fore-closed federal partisan-gerrymandering review on the ground that no judicially manageable standard exists for distinguishing constitutionally acceptable from unacceptable partisan motivation in line-drawing. The Court’s reasoning supposes a line-drawer whose motivation is the analytic object. A DualBalance plan has no such object. State-court tests that rely on the Efficiency Gap or mean-median asymmetry typically interpret a non-zero value as *prima facie* evidence of partisan intent; on a DualBalance plan the same value is evidence only of a particular voter-density distribution. The plan is in no useful sense “partisan”; it is a function of geometry and population, evaluated against a layer of vote data the function never saw.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [6, 8] compare an enacted plan to a sample of legally compliant alternatives that a hypothetical neutral line-drawer might have produced. The ensemble is meaningful because it stands in for the absent neutral line-drawer; the enacted plan being an outlier suggests the actual line-drawer differed from the neutral one. DualBalance *is* the neutral line-drawer, in a much stronger sense than the ensemble’s sample mean. Asking whether the DualBalance plan is an outlier with respect to its own ensemble is not an obviously meaningful question; the ensemble has been collapsed to a single point.

These observations do not amount to a claim that metrics should be ignored. They are useful as descriptive geometry, as inputs to a public discussion of what a plan looks like, and as comparison anchors when the same metrics are applied to several plans together. What they cannot do, on a deterministic generated plan, is what they were built to do on an enacted plan: tell us whether a person manipulated the lines and to what end.

4.8 What this paper claims, and what it does not

The paper makes four claims.

First, that a knob-free, race-blind, partisan-blind, deterministic districting algorithm exists that produces output competitive with, and on a single empirically tested state superior to, the corresponding hand-drawn enacted plan on a prespecified population-and-area objective. The algorithm’s output is byte-reproducible and updates only with the ten-year census.

Second, that the algorithm’s procedural neutrality is symmetric. By construction, the rule cannot be used to harm any group, party, or incumbent, and cannot be used to help any of them either. The same property that makes a partisan gerrymander impossible makes a race-conscious remedy impossible. Whether the symmetry is morally preferable to the status quo of partisan and remedial discretion is a question the algorithm itself cannot answer; it is a question for legislatures, courts, and voters. We claim only that the symmetry is the algorithm’s defining property, not an incidental side-effect.

Third, that the post-*Callais* legal landscape gives a generator of this kind a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*; state-court remedies for partisan claims depend on the politics of each state’s supreme court. A rule whose inputs do not contain race or party is, in the strict sense, not subject to either body of doctrine.

Fourth, that the conventional gerrymandering metrics (compactness scores, Efficiency Gap, mean-median, ensemble outliers, majority-minority counts) are forensic instruments for inferring intent in a line-drawing process that has none under this procedure. They continue to compute and are reported alongside the DualBalance metrics for transparency; they should not be taken to license, on this plan, the inferences they license on an enacted plan. This is the strongest framing claim of the paper, and the one most likely to draw resistance: it asks the redistricting literature to accept that an entire toolkit of normative measurement, developed over thirty years for very good reasons, is the wrong toolkit for the generative case.

The paper does not claim that the DualBalance Score is universally the right objective for redistricting, that radial slicing dominates all enacted plans across all states, that compactness can or should be ignored, or that Section 2 of the Voting Rights Act can be safely disregarded after *Callais*. The trade-offs documented above are real, and a serious adopter of the method would have to weigh them against the trade-offs the status quo presents.

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